

E ON THE BEST CURVATURE SPLITTING PROBLEM (see Section 4)

In Section 4 we examined how to achieve the optimal splitting for a given objective $F = f_1 - f_2$ by subtracting the same curvature term $\lambda \frac{\|\cdot\|^2}{2}$ in both functions. We concluded that if $\mu_1 \leq \mu_2$, the best splitting is achieved by shifting the lower curvature of f_1 to 0, i.e., make it convex. Conversely, when $\mu_1 > \mu_2$, the best splitting is achieved for some f_2 weakly convex, with lower curvature $\mu_2 - \lambda < 0$.

Within this section, we provide additional numerical experiments. In Figure 10 we provide an example with fixed curvature bounds of a nonconvex-nonconcave objective function F , namely $\mu_F = -0.5$ and $L_F = 1.5$, and illustrate all possible regimes after one iteration. Note that $\mu_F = \mu_1 - L_2$ and $L_F = L_1 - \mu_2$; further on, we examine the regimes based on the ranges of L_2 and μ_2 . The condition $\mu_1 \geq 0$ implies that $L_2 = \mu_1 - \mu_F \geq -\mu_F$, while the condition $\mu_1 + \mu_2 > 0$ that $\mu_2 > -\mu_1 = L_2 + \mu_F$. The contour lines represent the values of the denominators p_i , with $i = 1, 2, 3, 4$. The **red** points mark the initial curvature values of L_2 and μ_2 , whereas the **green** dots indicate the points with the largest possible p_i obtained through the optimal choice of λ . Since these shifts are linear in λ , the dashed lines connecting the dots have a slope of one.

For example, in the case where $L_2 = 1$ and $\mu_2 = 0.75$, we have $\mu_1 < \mu_2$ and the best splitting is obtained by shifting to the lowest possible value of L_2 , corresponding to $\mu_1 = 0$. In all other examples, $\mu_1 > \mu_2$ and the optimal splittings are found within regime p_3 , where $\mu_2 < 0$.

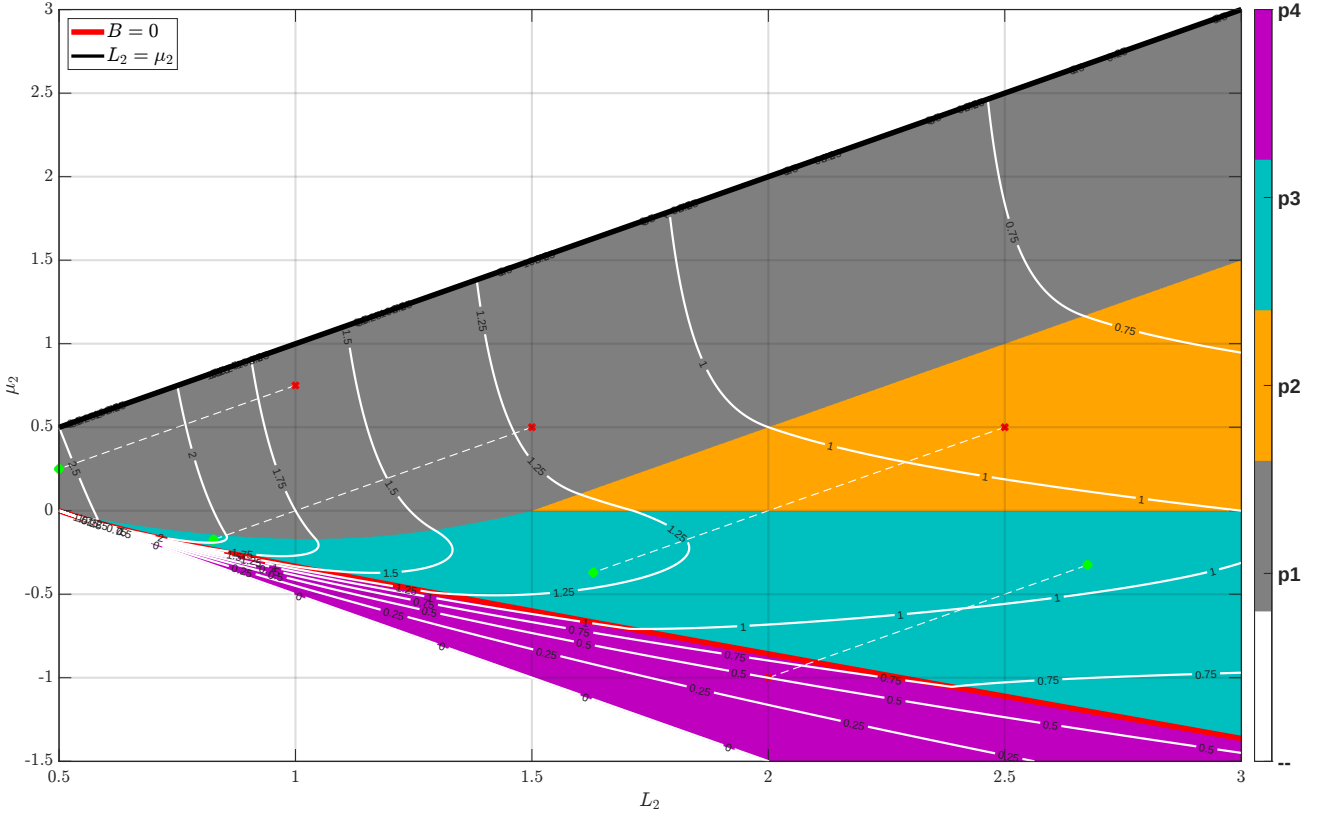


Figure 10: All regimes for a fixed objective function F with $\mu_F = -0.5$ and $L_F = 1.5$, along with several mappings of the optimal splittings, shown as transitions from **red** to **green** dots along dashed lines with a slope of 1.